

```
int water;
```

```
int trigpin=D1; int echopin=D2; long duration; int distance;
```

```
int BUZZER=D4;
```

```
int pirPin = D5; int ledPin = D6;
```

```
int buzzerPin = D4; int relayPin =D3; int pirState = HIGH; int val = 1;
```

```
//Blynk Virtual Pins #define VPIN_SOIL V0
```

```
#define VPIN_DISTANCE V1 #define VPIN_PIR V2
```

```
#define VPIN_PUMP_CONTROL V3 #define VPIN_BUZZER V4
```

```
// Blynk Function to Control Pump Manually
```

```
/*BLYNK_WRITE(VPIN_PUMP_CONTROL)
```

```
{
```

```
int pinValue = param.asInt(); // Get value from app button digitalWrite(relayPin, pinValue);
```

```
Serial.print("Pump manually set to: "); Serial.println(pinValue ? "ON" : "OFF");
```

```
*/
```

```
void setup() { pinMode(D3,OUTPUT); pinMode(D0,INPUT); pinMode(trigpin,OUTPUT);
```

```
pinMode(echopin,INPUT); pinMode(BUZZER,OUTPUT);
```

```
pinMode(pirPin, INPUT); pinMode(ledPin, OUTPUT); pinMode(buzzerPin, OUTPUT);
```

```
Serial.println("PIR Motion Sensor Test Starting..."); Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
water = digitalRead(D0); if(water == HIGH)
```

```
{
```

```
digitalWrite(D3,LOW);
```

```
}
```

```
else
```

```
{
```

```
digitalWrite(D3,HIGH);
```

```
}delay(400);
```

```
digitalWrite(trigpin,LOW);
```

```

delayMicroseconds(2);
digitalWrite(trigpin,HIGH);
delayMicroseconds(20);
digitalWrite(trigpin,LOW);
duration=pulseIn(echopin,HIGH);
distance=duration*0.034/2;
Serial.print("DISTANCE IN CM");
Serial.println(distance);
if ((distance<=100))
{
digitalWrite(BUZZER,HIGH);
Serial.println("OBJ ECT IS DETECTED");
}
else
{
digitalWrite(BUZZER,LOW);
Serial.println("OBJECT IS NOT DETECTED");
}
val = digitalRead(pirPin); if (val == HIGH)
{
digitalWrite(ledPin, HIGH); delay(1000);
if (pirState == LOW)
{
Serial.println("△ Motion detected!"); pirState = HIGH;
delay(1000);
}
}
else
{
digitalWrite(ledPin, LOW);
if (pirState == HIGH)

```

```
{  
  Serial.println("Motion ended."); pirState = LOW;  
}  
}  
}
```